LEARNING HUB MASTER SERIES • VOLUME 09

Linear Algebra: Singular Value Decomposition (SVD) & Pseudoinverse

The Pinnacle of Matrix Factorization: Geometry of SVD, The Four Fundamental Subspaces, Low-Rank Matrix Approximations (Eckart-Young), The Moore-Penrose Pseudoinverse, Minimum-Norm Least Squares, Matrix Norms, Condition Numbers, and Polar Decomposition

Synthesized from International Standard Curricula:
• **Primary Source:** Gilbert Strang — *Introduction to Linear Algebra* (5th Edition, Chapter 7: The SVD)
• **Visual Intuition:** Kenji Hiranabe — *The Art of Linear Algebra* (Pattern P5 & Fig. 19/20 SVD Maps)

Standards: Canonical Reference Edition • Stored in `Document/Linear-Algebra/EN/`
Key Features: 4 High-Clarity Vector SVGs • 6 Hand Calculation Fast-Track Shortcut Cards (S1–S6) • 5 Step-by-Step Worked Problems • 5 Examination Traps

Table of Contents

Part 1: The Foundations & Geometry of SVD

Chapter 1: The SVD Theorem — The Master Factorization of Any Matrix — Core Theory

Chapter 2: Geometric Transformation: Unit Sphere to Hyper-Ellipsoid — Figure 1 (SVG)

Chapter 3: SVD and the Four Fundamental Subspaces (The Big Picture) — Figure 2 (SVG)

Part 2: Matrix Approximations & Generalized Inverses

Chapter 4: Rank-1 Summation & Eckart-Young Low-Rank Approximations — Figure 3 (SVG)

Chapter 5: The Moore-Penrose Pseudoinverse & Minimum-Norm Least Squares — Figure 4 (SVG)

Chapter 6: Matrix Norms, Condition Number, and Polar Decomposition — Applications

Part 3: Hand Calculation Fast-Track Shortcut Cards

TRICK S1: The A^TA vs. AA^T Size Selection Shortcut — Speedup 5×

TRICK S2: Outer-Product Rank-1 SVD by Pure Inspection — Instant

TRICK S3: Computing Left Singular Vectors via $u_i = (1/\sigma_i)Av_i$ — Zero AA^T

TRICK S4: Instant Pseudoinverse Formula $A^+ = V\,\Sigma^+\,U^T$ — Formula

TRICK S5: Rapid Matrix Compression Error Calculation via σ_{k+1} — 10 Seconds

TRICK S6: Reading Four Subspaces Directly from U and V — Inspection

Part 4: Step-by-Step Worked Problems

Problem 9.1: Full SVD of a 2×3 Rectangular Matrix (Detailed Verification) — Rectangular

Problem 9.2: SVD of a 3×2 Tall-Thin Matrix & Left Nullspace Vector — Tall Matrix

Problem 9.3: SVD and Pseudoinverse of a Rank-1 Outer Product Matrix — Rank-1

Problem 9.4: Eckart-Young Best Rank-1 Approximation & Energy Fraction — Compression

Problem 9.5: Minimum-Norm Solution of an Underdetermined Linear System — Optimization

Part 5: Common Pitfalls & Exam Traps

5 Classic Mistakes: Sorting σ_i, Matrix Dimensions, Unit Norms, Sign Sync, Big Matrices — Caution

Part 1: The Foundations & Geometry of SVD

Chapter 1: The SVD Theorem — The Master Factorization of Any Matrix

In the study of Linear Algebra, diagonalization $A = X\Lambda X^{-1}$ represents a tremendous triumph for square matrices. However, it suffers from two fatal limitations: (1) it requires A to be square ($m = n$), and (2) it requires A to possess a complete set of n linearly independent eigenvectors (failing for defective matrices). The **Singular Value Decomposition (SVD)** removes every restriction: it decomposes **any real matrix of any dimensions $m \times n$** into orthogonal matrices and a non-negative diagonal scaling matrix.

▶ **The Singular Value Decomposition Theorem**

Every real matrix $A \in \mathbb{R}^{m \times n}$ can be factored into the product of three fundamental matrices:

$$A = U \Sigma V^T$$

where:

- $U \in \mathbb{R}^{m \times m}$ is an **Orthogonal Matrix** ($U^T U = I_m$). The columns $\mathbf{u}_1, \ldots, \mathbf{u}_m$ are called the **Left Singular Vectors** (orthonormal eigenvectors of AA^T).
- $V \in \mathbb{R}^{n \times n}$ is an **Orthogonal Matrix** ($V^T V = I_n$). The columns $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are called the **Right Singular Vectors** (orthonormal eigenvectors of $A^T A$).
- $\Sigma \in \mathbb{R}^{m \times n}$ is an $m \times n$ diagonal matrix containing the **Singular Values** on its main diagonal, sorted in non-increasing order:

$$\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0, \quad \sigma_{r+1} = \ldots = \sigma_{\min(m,n)} = 0$$

where $r = \text{rank}(A)$.

1.1 The Origin of Singular Values from Symmetric Matrices A^TA and AA^T

To understand where U, Σ, and V come from, we examine the two symmetric, positive semidefinite Gram matrices $A^TA \in \mathbb{R}^{n \times n}$ and $AA^T \in \mathbb{R}^{m \times m}$. Substituting $A = U\Sigma V^T$ into both products yields:

$$A^TA = (U\Sigma V^T)^T (U\Sigma V^T) = V \Sigma^T U^T U \Sigma V^T = V (\Sigma^T \Sigma) V^T$$

$$AA^T = (U\Sigma V^T) (U\Sigma V^T)^T = U \Sigma V^T V \Sigma^T U^T = U (\Sigma \Sigma^T) U^T$$

Since A^TA is symmetric and positive semidefinite, its eigenvalues λ_i are non-negative real numbers. Thus:

- The non-zero eigenvalues of A^TA and AA^T are **identical** and equal to σ_i^2:

$$\sigma_i = \sqrt{(\lambda_i(A^TA))} = \sqrt{(\lambda_i(AA^T))}$$

- The columns of V are the orthonormal eigenvectors of A^TA: $(A^TA)\mathbf{v}_i = \sigma_i^2\,\mathbf{v}_i$.
- The columns of U are the orthonormal eigenvectors of AA^T: $(AA^T)\mathbf{u}_i = \sigma_i^2\,\mathbf{u}_i$.

▶ The Core Fundamental Mapping Relation: $A\mathbf{v}_i = \sigma_i\,\mathbf{u}_i$

Multiplying A by $\mathbf{v}_i$ directly produces:

$$A\mathbf{v}_i = (U\Sigma V^T)\,\mathbf{v}_i = U\,\Sigma\,\mathbf{e}_i = U\,(\sigma_i\,\mathbf{e}_i) = \sigma_i\,\mathbf{u}_i \qquad (\text{for } 1 \le i \le r)$$

For $i > r$ (where $\sigma_i = 0$), we have $A\mathbf{v}_i = \mathbf{0}$. This establishes the critical operational formula:

$$\mathbf{u}_i = \frac{1}{\sigma_i} A\mathbf{v}_i \qquad (1 \le i \le r)$$

Chapter 2: Geometric Transformation: Unit Sphere to Hyper-Ellipsoid

Geometrically, every linear transformation $A \in \mathbb{R}^{m \times n}$ maps the unit sphere $S^{n-1} = \{\,\mathbf{x} \in \mathbb{R}^n : ||\mathbf{x}|| = 1\,\}$ into a **hyper-ellipsoid** in $\mathbb{R}^m$. The SVD reveals the exact three-stage breakdown of this geometric mapping:

1. **First Rotation / Reflection (V^T):** Rotates the domain $\mathbb{R}^n$ so that the principal axes $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ align with the standard coordinate axes.
2. **Diagonal Scaling (Σ):** Stretches the coordinate axes by factor σ_i for $i \le r$, and completely flattens (collapses) the remaining $n - r$ directions to zero.
3. **Second Rotation / Reflection (U):** Rotates the scaled axes in the codomain $\mathbb{R}^m$ so that the principal semi-axes point along $\{\mathbf{u}_1, \ldots, \mathbf{u}_r\}$.

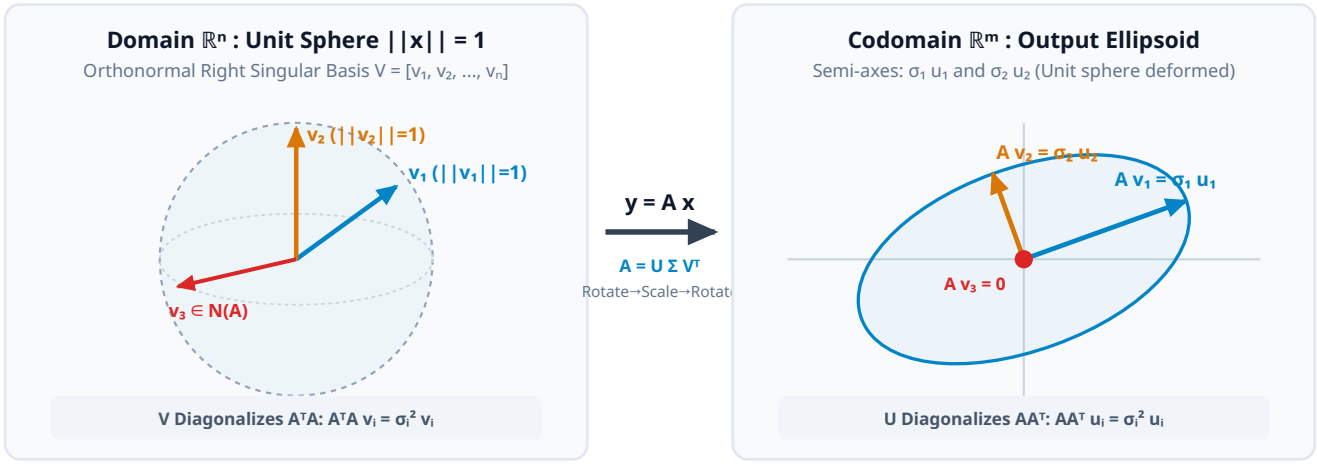


Figure 1: Geometric mapping of SVD: An orthogonal triad of right singular vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ on the unit sphere in $\mathbb{R}^3$ is mapped to the semi-axes of an ellipsoid in $\mathbb{R}^2$ with lengths σ_1 and σ_2, while $\mathbf{v}_3 \in N(A)$ is collapsed to 0.

Chapter 3: SVD and the Four Fundamental Subspaces (The Big Picture)

The SVD provides the most complete and elegant realization of Gilbert Strang's **Four Fundamental Subspaces**. The columns of V and U provide tailored, orthonormal bases for all four spaces simultaneously:

▶ The Orthonormal Bases of the Four Fundamental Subspaces

Let $A = U\Sigma V^T$ have rank r. Then:

- **Row Space $C(A^T) \subset \mathbb{R}^n$:** dim = r. Orthonormal basis: $\{ v_1, v_2, ..., v_r \}$.
- **Nullspace $N(A) \subset \mathbb{R}^n$:** dim = $n - r$. Orthonormal basis: $\{ v_{r+1}, ..., v_n \}$.
- **Column Space $C(A) \subset \mathbb{R}^m$:** dim = r. Orthonormal basis: $\{ u_1, u_2, ..., u_r \}$.
- **Left Nullspace $N(A^T) \subset \mathbb{R}^m$:** dim = $m - r$. Orthonormal basis: $\{ u_{r+1}, ..., u_m \}$.

Orthogonality holds unconditionally: $C(A^T) \perp N(A)$ in $\mathbb{R}^n$, and $C(A) \perp N(A^T)$ in $\mathbb{R}^m$.

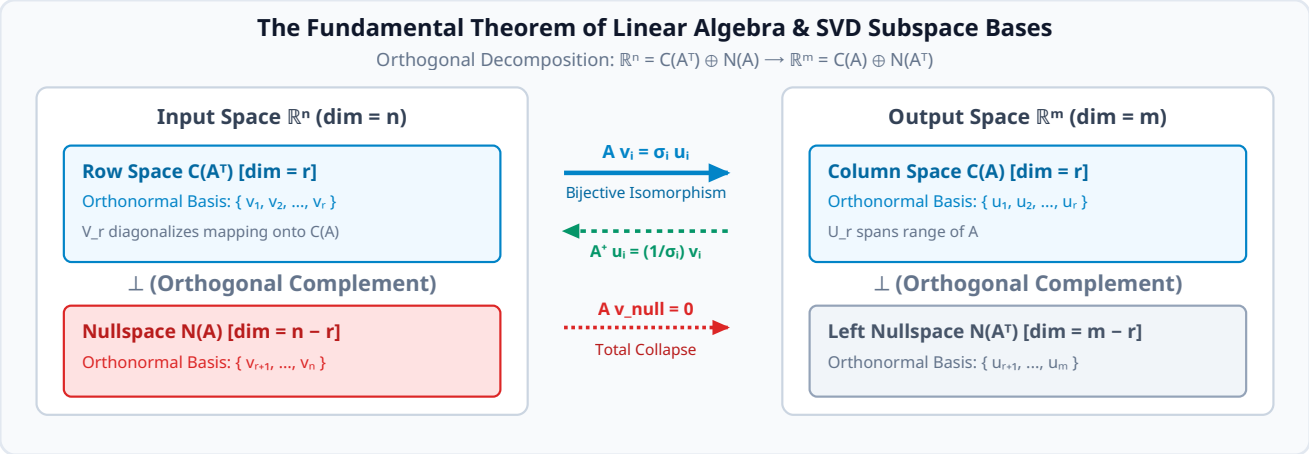


Figure 2: The SVD Big Picture: Orthogonal decomposition of domain $\mathbb{R}^n = C(A^T) \oplus N(A)$ and codomain $\mathbb{R}^m = C(A) \oplus N(A^T)$, showing that A acts as an invertible scaling isomorphism between the two r-dimensional spaces.

3.1 Full SVD vs. Compact (Reduced) SVD

Depending on whether the nullspaces are included, the SVD can be written in two standard forms:

- **Full SVD:** $A = U \Sigma V^T$ where U is $m \times m$, Σ is $m \times n$, and V is $n \times n$. This form includes bases for the nullspaces $N(A)$ and $N(A^T)$.
- **Compact (Reduced) SVD:** $A = U_r \Sigma_r V_r^T$ where U_r is $m \times r$, $\Sigma_r = \text{diag}(\sigma_1, ..., \sigma_r)$ is $r \times r$ invertible, and V_r is $n \times r$. This discards all zero singular values and nullspace components while reproducing the exact same matrix A.

Part 2: Matrix Approximations & Generalized Inverses

Chapter 4: Rank-1 Summation & Eckart-Young Low-Rank Approximations

By expanding the block matrix product $U_r \Sigma_r V_r^T$ as outer products of columns, any matrix A can be expressed as a linear combination of r rank-1 matrices:

$$A = \sum_{i=1}^{r} \sigma_i \mathbf{u}_i \mathbf{v}_i^T = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T + \sigma_2 \mathbf{u}_2 \mathbf{v}_2^T + \dots + \sigma_r \mathbf{u}_r \mathbf{v}_r^T$$

▶ The Eckart-Young-Mirsky Low-Rank Approximation Theorem

For any integer k with $1 \le k < r$, the optimal rank-k approximation to A that minimizes the error in both the induced 2-norm and Frobenius norm is obtained by truncating the sum at k terms:

$$A_k = \sum_{i=1}^{k} \sigma_i \mathbf{u}_i \mathbf{v}_i^T$$

The minimal approximation errors are exactly:

$$\min_{\text{rank}(B) \le k} ||A - B||_2 = ||A - A_k||_2 = \sigma_{k+1}$$

$$\min_{\text{rank}(B) \le k} ||A - B||_F^2 = ||A - A_k||_F^2 = \sum_{j=k+1}^{r} \sigma_j^2$$

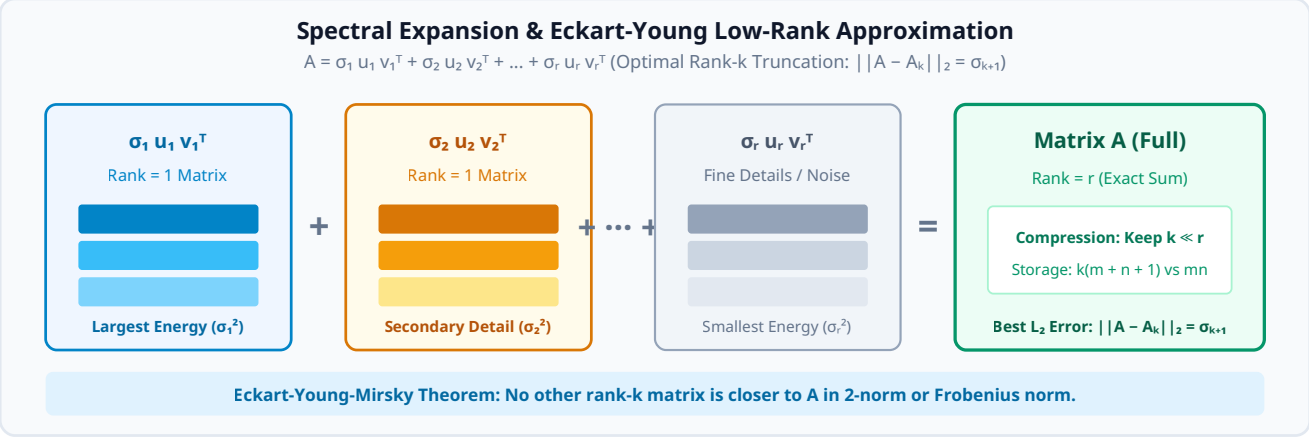


Figure 3: Spectral expansion: Matrix A is the sum of r rank-1 matrices weighted by singular values σ_i. Retaining only the first k dominant terms yields the optimal compressed matrix A_k.

Chapter 5: The Moore-Penrose Pseudoinverse & Minimum-Norm Least Squares

When a matrix A is rectangular or singular, the standard inverse A^{-1} does not exist. The **Moore-Penrose Pseudoinverse**, denoted $A^{+} \in \mathbb{R}^{n \times m}$, provides a universal generalized inverse defined via the SVD.

▶ Definition of the Moore-Penrose Pseudoinverse

If $A = U\Sigma V^{T}$ is the Full SVD of A, its pseudoinverse is:

$$A^{+} = V \Sigma^{+} U^{T} = \sum_{i=1}^{r} \frac{1}{\sigma_i} \mathbf{v}_i \mathbf{u}_i^{T}$$

where $\Sigma^{+} \in \mathbb{R}^{n \times m}$ is formed by transposing Σ and replacing each non-zero singular value σ_i with its reciprocal $1/\sigma_i$.

A^{+} is the unique matrix satisfying the **Four Penrose Equations**:

$$1)\ AA^{+}A = A, \quad 2)\ A^{+}AA^{+} = A^{+}, \quad 3)\ (AA^{+})^{T} = AA^{+}, \quad 4)\ (A^{+}A)^{T} = A^{+}A$$

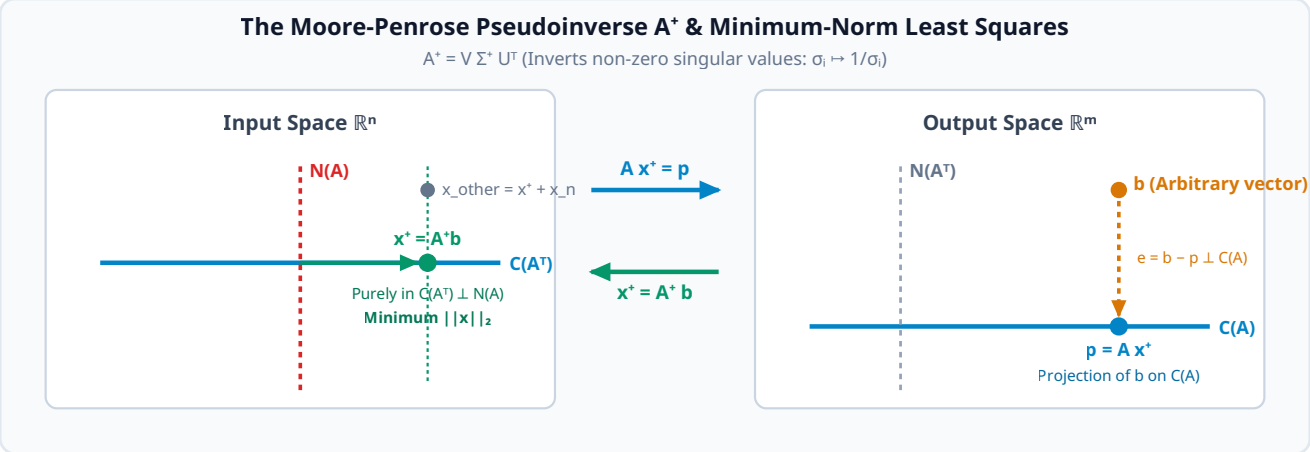


Figure 4: The Moore-Penrose Pseudoinverse geometry: For an inconsistent or underdetermined system Ax = b, the vector $x^{+} = A^{+}b$ provides the unique solution that projects b onto C(A) and lies entirely within $C(A^{T})$, guaranteeing minimum Euclidean norm.

5.1 Solving Arbitrary Systems Ax = b with A^{+}

For any linear system $A\mathbf{x} = \mathbf{b}$, the pseudoinverse solution $\mathbf{x}^{+} = A^{+}\mathbf{b}$ resolves both fundamental pathologies of linear systems:

1. **When no solution exists (Inconsistent System): $\mathbf{x}^{+}$** is the **Least Squares Solution** that minimizes the residual error $||A\mathbf{x} - \mathbf{b}||_2$. In this case, $A\mathbf{x}^{+} = \mathbf{p} = P_{C(A)}\,\mathbf{b}$.

2. **When infinitely many solutions exist (Underdetermined System):** $\mathbf{x}^+$ selects the **unique solution with minimal Euclidean norm** $||\mathbf{x}||_2$. This occurs because $\mathbf{x}^+ \in C(A^T)$, containing zero component in the nullspace $N(A)$.

Chapter 6: Matrix Norms, Condition Number, and Polar Decomposition

Singular values define the most important analytical metrics of a matrix in numerical computation:

- **Induced 2-Norm:** The maximum amplification factor of A is its largest singular value:

$$||A||_2 = \max_{||\mathbf{x}||=1} ||A\mathbf{x}|| = \sigma_1 = \sigma_{max}$$

- **Frobenius Norm:** $||A||_F = \sqrt{(\mathrm{tr}(A^T A))} = \sqrt{(\sum_{i=1}^{r} \sigma_i^2)}$.
- **Condition Number:** Measures sensitivity to perturbations in $A\mathbf{x} = \mathbf{b}$:

$$\kappa(A) = ||A||_2 \, ||A^+||_2 = \text{" + frac("}\sigma_{max}\text{", "}\sigma_{min}\text{") + " = " + frac("}\sigma_1\text{", "}\sigma_r\text{") + " } \geq 1$$

  A matrix with $\kappa(A) \approx 1$ is **well-conditioned**; when $\kappa(A) \gg 1$, it is **ill-conditioned**.
- **Polar Decomposition:** Any square real matrix A can be factored into an orthogonal matrix Q (rotation/reflection) and a positive semidefinite symmetric matrix S:

$$A = QS \quad \text{with} \quad Q = UV^T \quad \text{and} \quad S = V\Sigma V^T = \sqrt{(A^T A)}$$

Part 3: Hand Calculation Fast-Track Shortcut Cards

Computing the SVD by hand can be laborious if one blindly follows brute-force matrix multiplication. The following 6 Fast-Track shortcut cards (S1–S6) streamline hand calculations for examination scenarios:

TRICK S1 The A^TA vs. AA^T Size Selection Shortcut

Saves 90% Time

Standard Method: Always computing A^TA regardless of matrix shape. For a 2×100 matrix, this forces the student to compute and diagonalize a 100×100 matrix!

⚡ **Fast Hand Calculation:** Always compute the **smaller** matrix! If $m < n$, compute the $m \times m$ matrix AA^T to find $\sigma_1, \ldots, \sigma_r$ and $\mathbf{u}_1, \ldots, \mathbf{u}_r$. Then find $\mathbf{v}_i$ via $\mathbf{v}_i = (1/\sigma_i)\, A^T\mathbf{u}_i$. If $n < m$, compute A^TA.

TRICK S2 Outer-Product Rank-1 SVD by Pure Inspection

Instant Inspection

Standard Method: Setting up characteristic polynomials $\det(A^TA - \lambda I) = 0$ for outer products $A = \mathbf{x}\,\mathbf{y}^T$.

⚡ **Fast Hand Calculation:** For any rank-1 outer product $A = \mathbf{x}\,\mathbf{y}^T$, there is only **one non-zero singular value**: $\sigma_1 = ||\mathbf{x}|| \cdot ||\mathbf{y}||$. The left and right singular vectors are the normalized vectors: $\mathbf{u}_1 = \mathbf{x} / ||\mathbf{x}||$ and $\mathbf{v}_1 = \mathbf{y} / ||\mathbf{y}||$. All other singular values $\sigma_2 = \ldots = 0$. SVD is found in 5 seconds!

TRICK S3 Computing Left Singular Vectors via $\mathbf{u_i} = (1/\sigma_i)\, A\, \mathbf{v_i}$

Zero Extra Eigenspaces

Standard Method: Solving the second eigenvalue problem $\det(AA^T - \lambda I) = 0$ from scratch, risking sign inconsistencies.

⚡ **Fast Hand Calculation:** Never diagonalize both A^TA and AA^T! Once $\mathbf{v}_i$ is found from A^TA, compute $\mathbf{u}_i$ directly via $\mathbf{u}_i = \dfrac{1}{\sigma_i} A\mathbf{v}_i$. This guarantees correct signs that satisfy $A = U\Sigma V^T$.

TRICK S4 Instant Pseudoinverse from Compact SVD

One-Line Formula

Standard Method: Inverting large products or setting up normal equations $(A^TA)^{-1}A^T$.

⚡ **Fast Hand Calculation:** From the compact SVD $A = \sum_{i=1}^{r} \sigma_i\, \mathbf{u}_i\, \mathbf{v}_i^T$, write A^+ immediately by flipping vectors and inverting σ_i: $A^+ = \sum_{i=1}^{r} \dfrac{1}{\sigma_i}\, \mathbf{v}_i\, \mathbf{u}_i^T = V_r\, \Sigma_r^{-1}\, U_r^T$.

TRICK S5 Rapid Matrix Compression Error via σ_{k+1}

Read Off Instantly

Standard Method: Constructing the truncated matrix A_k, subtracting $A - A_k$, and calculating the induced norm of the difference matrix.

⚡ **Fast Hand Calculation:** By Eckart-Young, the 2-norm error is **identically equal to the first neglected singular value**: $||A - A_k||_2 = \sigma_{k+1}$. The relative Frobenius energy captured is exactly $\sum_{i=1}^{k} \sigma_i^2 / \sum_{i=1}^{r} \sigma_i^2$.

TRICK S6 Reading the Four Fundamental Subspaces Directly from U and V

Zero Elimination Needed

Standard Method: Running separate row eliminations (rref) to find pivot and free columns.

⚡ **Fast Hand Calculation:** In the Full SVD $A = U\Sigma V^T$ with rank r: First r cols of V span $C(A^T)$; last $n - r$ cols of V span $N(A)$; first r cols of U span $C(A)$; last $m - r$ cols of U span $N(A^T)$. All bases are automatically orthonormal!

Part 4: Step-by-Step Worked Problems

Problem 9.1 **Full SVD of a 2×3 Rectangular Matrix**

Find the Full Singular Value Decomposition $A = U\,\Sigma\,V^T$ for the matrix

$$A = \begin{vmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{vmatrix}$$

Verify the product $U\Sigma V^T = A$, and identify orthonormal bases for all four fundamental subspaces.

STEP-BY-STEP SOLUTION **Full SVD Construction for 2×3 Matrix**

Step 1: Size Decision — Compute the 2×2 Matrix AA^T (Fast-Track S1)

Since A is 2×3 ($m = 2 < n = 3$), computing AA^T (2×2) is far faster than computing A^TA (3×3):

$$AA^T = \begin{vmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{vmatrix} \begin{vmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{vmatrix} = \begin{vmatrix} 9+4+4 & 6+6-4 \\ 6+6-4 & 4+9+4 \end{vmatrix} = \begin{vmatrix} 17 & 8 \\ 8 & 17 \end{vmatrix}$$

Step 2: Eigenvalues and Left Singular Vectors (U)

The characteristic equation of AA^T is:

$$\lambda^2 - \text{tr}(AA^T)\lambda + \det(AA^T) = \lambda^2 - 34\lambda + (289 - 64) = \lambda^2 - 34\lambda + 225 = (\lambda - 25)(\lambda - 9) = 0$$

Sorting in decreasing order: $\boldsymbol{\lambda_1 = 25}$ and $\boldsymbol{\lambda_2 = 9}$. Singular values are:

$$\sigma_1 = \sqrt{25} = 5, \qquad \sigma_2 = \sqrt{9} = 3$$

Now find the orthonormal eigenvectors $\mathbf{u}_1, \mathbf{u}_2$ of AA^T:

- For $\lambda_1 = 25$: $(AA^T - 25I)\mathbf{u}_1 = \begin{vmatrix} -8 & 8 \\ 8 & -8 \end{vmatrix} \mathbf{u}_1 = \mathbf{0} \Rightarrow \mathbf{u}_1 = \frac{1}{\sqrt{2}} \begin{vmatrix} 1 \\ 1 \end{vmatrix}$
- For $\lambda_2 = 9$: $(AA^T - 9I)\mathbf{u}_2 = \begin{vmatrix} 8 & 8 \\ 8 & 8 \end{vmatrix} \mathbf{u}_2 = \mathbf{0} \Rightarrow \mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{vmatrix} 1 \\ -1 \end{vmatrix}$

$$U = \begin{vmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{vmatrix}$$

Step 3: Right Singular Vectors (V) via $\mathbf{v}_i = (1/\sigma_i)\, A^T \mathbf{u}_i$ (Fast-Track S3)

We compute $\mathbf{v}_1$ and $\mathbf{v}_2$ directly from $A^T\mathbf{u}_i$:

$$\mathbf{v}_1 = \frac{1}{5}\begin{vmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{vmatrix} \frac{1}{\sqrt{2}}\begin{vmatrix} 1 \\ 1 \end{vmatrix} = \frac{1}{5\sqrt{2}}\begin{vmatrix} 5 \\ 5 \\ 0 \end{vmatrix} = \frac{1}{\sqrt{2}}\begin{vmatrix} 1 \\ 1 \\ 0 \end{vmatrix}$$

$$\mathbf{v}_2 = \frac{1}{3}\begin{vmatrix} 3 & 2 \\ 2 & 3 \\ 2 & -2 \end{vmatrix} \frac{1}{\sqrt{2}}\begin{vmatrix} 1 \\ -1 \end{vmatrix} = \frac{1}{3\sqrt{2}}\begin{vmatrix} 1 \\ -1 \\ 4 \end{vmatrix} = \frac{1}{3\sqrt{2}}\begin{vmatrix} 1 \\ -1 \\ 4 \end{vmatrix}$$

Step 4: Nullspace Vector $\mathbf{v}_3 \in N(A)$

The third vector $\mathbf{v}_3$ spans $N(A)$ and must be orthogonal to both $\mathbf{v}_1$ and $\mathbf{v}_2$. Taking the cross product of $[1, 1, 0]^T$ and $[1, -1, 4]^T$:

$$[1, 1, 0]^T \times [1, -1, 4]^T = [4, -4, -2]^T = 2\,[2, -2, -1]^T$$

Normalizing by $\sqrt{(4 + 4 + 1)} = 3$ gives:

$$\mathbf{v}_3 = \frac{1}{3}\begin{vmatrix} 2 \\ -2 \\ -1 \end{vmatrix}$$

Step 5: Assemble Full SVD and Verify Product

$$\Sigma = \begin{vmatrix} 5 & 0 & 0 \\ 0 & 3 & 0 \end{vmatrix}$$

$$V = \begin{vmatrix} 1/\sqrt{2} & 1/(3\sqrt{2}) & 2/3 \\ 1/\sqrt{2} & -1/(3\sqrt{2}) & -2/3 \\ 0 & 4/(3\sqrt{2}) & -1/3 \end{vmatrix}, \quad V^T = \begin{vmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/(3\sqrt{2}) & -1/(3\sqrt{2}) & 4/(3\sqrt{2}) \\ 2/3 & -2/3 & -1/3 \end{vmatrix}$$

Verification: Computing ΣV^T and multiplying by U:

$$\Sigma\, V^T = \begin{vmatrix} 5/\sqrt{2} & 5/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 4/\sqrt{2} \end{vmatrix}$$

$$U(\Sigma V^T) = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 5/\sqrt{2} & 5/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 4/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix} = A \quad \checkmark$$

Verified!

Problem 9.2 **Full SVD of a 3×2 Tall-Thin Matrix & Left Nullspace Vector**

Compute the Full SVD of the 3 × 2 matrix

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$$

Explicitly construct the Left Nullspace vector $\mathbf{u}_3 \in N(A^T)$ to complete the 3 × 3 matrix U.

STEP-BY-STEP SOLUTION **Full SVD of Tall-Thin Matrix**

Step 1: Compute A^TA (2×2)

$$A^TA = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

Notice that A^TA is already diagonal! Eigenvalues in decreasing order are:

$$\lambda_1 = 3, \quad \lambda_2 = 2 \quad \Rightarrow \quad \sigma_1 = \sqrt{3}, \quad \sigma_2 = \sqrt{2}$$

The right singular vectors are the standard basis vectors:

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \Rightarrow \quad V = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Step 2: Compute Left Singular Vectors $\mathbf{u}_1, \mathbf{u}_2$

$$\mathbf{u}_1 = \frac{1}{\sqrt{3}} A\,\mathbf{v}_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\mathbf{u}_2 = \frac{1}{\sqrt{2}} A\,\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{vmatrix} 1 \\ 0 \\ -1 \end{vmatrix}$$

Step 3: Find Left Nullspace Vector $\mathbf{u}_3 \in N(A^T)$

We require $\mathbf{u}_3 \perp \mathbf{u}_1$ and $\mathbf{u}_3 \perp \mathbf{u}_2$. Taking the cross product of $[1, 1, 1]^T$ and $[1, 0, -1]^T$:

$$[1, 1, 1]^T \times [1, 0, -1]^T = [-1, 2, -1]^T$$

Normalizing: $\sqrt{(-1)^2 + 2^2 + (-1)^2} = \sqrt{6}$:

$$\mathbf{u}_3 = \frac{1}{\sqrt{6}} \begin{vmatrix} -1 \\ 2 \\ -1 \end{vmatrix}$$

Step 4: Final Full SVD Representation

$$U = \begin{vmatrix} 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \end{vmatrix}, \quad \Sigma = \begin{vmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \\ 0 & 0 \end{vmatrix}, \quad V^T = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix}$$

Verification:

$$U\Sigma = \begin{vmatrix} 1 & 1 \\ 1 & 0 \\ 1 & -1 \end{vmatrix} \quad \Rightarrow \quad (U\Sigma)\,V^T = \begin{vmatrix} 1 & 1 \\ 1 & 0 \\ 1 & -1 \end{vmatrix} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{vmatrix} = A \quad \checkmark \text{ Verified!}$$

Problem 9.3 **SVD and Pseudoinverse of a Rank-1 Outer Product Matrix**

Given the rank-1 matrix

$$A = \begin{vmatrix} 3 & 4 \\ 6 & 8 \end{vmatrix}$$

1) Use Fast-Track S2 to determine its SVD by pure inspection.

2) Find the Moore-Penrose Pseudoinverse A^+ and verify the Penrose condition $AA^+A = A$.

STEP-BY-STEP SOLUTION **Rank-1 Outer Product SVD & Pseudoinverse**

Part 1: SVD by Pure Inspection (Fast-Track S2)

Notice that A is the outer product of $\mathbf{x} = [1, 2]^T$ and $\mathbf{y} = [3, 4]^T$:

$$A = \begin{vmatrix} 1 \\ 2 \end{vmatrix} \begin{vmatrix} 3 & 4 \end{vmatrix}$$

Calculating norms:

$$||\mathbf{x}|| = \sqrt{(1 + 4)} = \sqrt{5}, \qquad ||\mathbf{y}|| = \sqrt{(9 + 16)} = 5$$

Therefore, the unique non-zero singular value is:

$$\sigma_1 = ||\mathbf{x}|| \cdot ||\mathbf{y}|| = 5\sqrt{5}, \qquad \sigma_2 = 0$$

The unit singular vectors are simply the normalized vectors:

$$\mathbf{u}_1 = \frac{1}{\sqrt{5}} \begin{vmatrix} 1 \\ 2 \end{vmatrix}, \qquad \mathbf{v}_1 = \frac{1}{5} \begin{vmatrix} 3 \\ 4 \end{vmatrix}$$

Completing orthonormal bases with perpendicular unit vectors:

$$\mathbf{u}_2 = \frac{1}{\sqrt{5}} \begin{vmatrix} -2 \\ 1 \end{vmatrix}, \qquad \mathbf{v}_2 = \frac{1}{5} \begin{vmatrix} -4 \\ 3 \end{vmatrix}$$

Part 2: Computing the Pseudoinverse A^+ (Fast-Track S4)

Using the compact formula $A^+ = (1/\sigma_1)\, \mathbf{v}_1\, \mathbf{u}_1^T$:

$$A^+ = \frac{1}{5\sqrt{5}} \left(\frac{1}{5} \begin{vmatrix} 3 \\ 4 \end{vmatrix} \right) \left(\frac{1}{\sqrt{5}} \begin{vmatrix} 1 & 2 \end{vmatrix} \right) = \frac{1}{125} \begin{vmatrix} 3 \\ 4 \end{vmatrix} \begin{vmatrix} 1 & 2 \end{vmatrix} = \frac{1}{125} \begin{vmatrix} 3 & 6 \\ 4 & 8 \end{vmatrix}$$

Verification of Penrose Condition 1 ($AA^+A = A$):

$$AA^+ = \begin{vmatrix} 3 & 4 \\ 6 & 8 \end{vmatrix} \frac{1}{125} \begin{vmatrix} 3 & 6 \\ 4 & 8 \end{vmatrix} = \frac{1}{125} \begin{vmatrix} 25 & 50 \\ 50 & 100 \end{vmatrix} = \frac{1}{5} \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix}$$

$$(AA^+)\,A = \frac{1}{5} \begin{vmatrix} 1 & 2 \\ 2 & 4 \end{vmatrix} \begin{vmatrix} 3 & 4 \\ 6 & 8 \end{vmatrix} = \frac{1}{5} \begin{vmatrix} 15 & 20 \\ 30 & 40 \end{vmatrix} = \begin{vmatrix} 3 & 4 \\ 6 & 8 \end{vmatrix} = A \quad \checkmark \text{ Verified!}$$

Problem 9.4 **Eckart-Young Best Rank-1 Approximation & Energy Fraction**

A matrix $A \in \mathbb{R}^{3 \times 3}$ has singular values $\sigma_1 = 10$, $\sigma_2 = 3$, $\sigma_3 = 1$, with corresponding singular vectors $\mathbf{u}_1 = [1, 0, 0]^T$ and $\mathbf{v}_1 = [0, 1, 0]^T$.

1) Construct the optimal rank-1 approximation A_1.

2) Find the exact approximation errors in both the 2-norm and Frobenius norm.

3) What fraction of total matrix energy is preserved in A_1?

STEP-BY-STEP SOLUTION **Eckart-Young Approximation & Energy Analysis**

1. Construction of Optimal Rank-1 Matrix A_1

By the Eckart-Young theorem, the best rank-1 approximation is the first spectral term:

$$A_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T = 10 \begin{vmatrix} 1 \\ 0 \\ 0 \end{vmatrix} \begin{vmatrix} 0 & 1 & 0 \end{vmatrix} = 10 \begin{vmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix} = \begin{vmatrix} 0 & 10 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{vmatrix}$$

2. Exact Error Calculations (Fast-Track S5)

- **2-Norm Error:** The induced 2-norm error is exactly the first dropped singular value:

$$||A - A_1||_2 = \sigma_2 = 3$$

- **Frobenius Norm Error:** The squared Frobenius error is the sum of squared dropped singular values:

$$||A - A_1||_F = \sqrt{(\sigma_2^2 + \sigma_3^2)} = \sqrt{(9 + 1)} = \sqrt{10} \approx 3.162$$

3. Energy Fraction Captured

Total matrix energy in the Frobenius norm is the sum of all squared singular values:

$$||A||_F^2 = \sigma_1^2 + \sigma_2^2 + \sigma_3^2 = 100 + 9 + 1 = 110$$

The energy captured by A_1 is $\sigma_1^2 = 100$. Thus, the preserved energy fraction is:

$$\text{Energy Ratio} = \frac{100}{110} = \frac{10}{11} \approx 90.91\%$$

Interpretation: A single rank-1 matrix captures over 90.9% of the total energy of A!

Problem 9.5 **Minimum-Norm Solution of an Underdetermined System**

Consider the underdetermined equation with infinitely many solutions:

$$x_1 + x_2 + x_3 = 6$$

1) Formulate this as $A\mathbf{x} = \mathbf{b}$ and determine the pseudoinverse A^+.

2) Find the minimum-norm solution $\mathbf{x}^+ = A^+\mathbf{b}$.

3) Prove geometrically and algebraically that $\mathbf{x}^+ \in C(A^T)$ and achieves strictly smaller norm than any other solution.

STEP-BY-STEP SOLUTION **Minimum-Norm Solution via Pseudoinverse**

1. Formulation and Pseudoinverse A^+

Here $A = [1, 1, 1] \in \mathbb{R}^{1 \times 3}$ and $\mathbf{b} = [6]$. A has full row rank ($r = m = 1$). Since A has full row rank, we use the direct formula $A^+ = A^T(AA^T)^{-1}$:

$$AA^T = [1, 1, 1]\,[1, 1, 1]^T = 1 + 1 + 1 = [3]$$

$$A^+ = \begin{vmatrix} 1 \\ 1 \\ 1 \end{vmatrix} [3]^{-1} = \frac{1}{3} \begin{vmatrix} 1 \\ 1 \\ 1 \end{vmatrix}$$

2. Minimum-Norm Solution $\mathbf{x}^+$

$$\mathbf{x}^+ = A^+\mathbf{b} = \frac{1}{3} \begin{vmatrix} 1 \\ 1 \\ 1 \end{vmatrix} [6] = \begin{vmatrix} 2 \\ 2 \\ 2 \end{vmatrix}$$

Check: $2 + 2 + 2 = 6$ ✔. The norm of this solution is:

$$||\mathbf{x}^+||_2 = \sqrt{(4 + 4 + 4)} = \sqrt{12} = 2\sqrt{3} \approx 3.464$$

3. Proof of Minimum Norm Property

Any solution to $x_1 + x_2 + x_3 = 6$ can be written as $\mathbf{x} = \mathbf{x}^+ + \mathbf{x}_n$, where $\mathbf{x}_n \in N(A)$ satisfies $A\mathbf{x}_n = \mathbf{0}$. Notice that $\mathbf{x}^+ = 2[1, 1, 1]^T$ lies entirely in the row space $C(A^T) = \text{span}\{[1, 1, 1]^T\}$.

By the Fundamental Theorem of Linear Algebra, $C(A^T) \perp N(A)$. Thus $\langle \mathbf{x}^+, \mathbf{x}_n \rangle = 0$. By the Pythagorean Theorem:

$$||\mathbf{x}||_2^2 = ||\mathbf{x}^+ + \mathbf{x}_n||_2^2 = ||\mathbf{x}^+||_2^2 + ||\mathbf{x}_n||_2^2 \geq ||\mathbf{x}^+||_2^2$$

Equality holds if and only if $\mathbf{x}_n = \mathbf{0}$. For example, the alternative solution $[6, 0, 0]^T$ has norm $6 > 3.464$. Thus, $\mathbf{x}^+$ is the uniquely minimal norm solution (Q.E.D.).

Part 5: Common Pitfalls & Exam Traps

The SVD is rich with geometric subtlety. Avoid these 5 classic examination traps:

⚠ Trap 1: Forgetting to Sort Singular Values in Decreasing Order

The Mistake: Writing singular values as $\sigma_1 = 2$, $\sigma_2 = 5$. By universal definition, singular values **must** be sorted in non-increasing order: $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0$. Violating this will swap the order of singular vectors in U and V, invalidating low-rank approximations.

⚠ Trap 2: Matrix Dimension Mismatches in Full SVD

The Mistake: For a 2×3 matrix, making U a 3×3 matrix or V a 2×2 matrix.

Rule: For $A \in \mathbb{R}^{m \times n}$: U is always $\boldsymbol{m \times m}$, Σ is always $\boldsymbol{m \times n}$, and V is always $\boldsymbol{n \times n}$ (so V^T is $n \times n$).

⚠ Trap 3: Forgetting to Normalize Singular Vectors

The Mistake: Inserting unnormalized eigenvectors like $[1, 1, 0]^T$ into V.

Rule: The columns of both U and V must be **unit vectors** ($||\mathbf{u}_i|| = 1$, $||\mathbf{v}_i|| = 1$). Always divide each vector by its Euclidean length $\sqrt{(\sum x_j^2)}$.

⚠ Trap 4: Inconsistent Sign Selection Between U and V

The Mistake: Independently choosing arbitrary eigenvectors for both A^TA and AA^T. If you flip the sign of $\mathbf{v}_i$ without flipping $\mathbf{u}_i$, the product $U\Sigma V^T$ will equal $-A$!

Rule: Always compute $\mathbf{v}_i$ first, then compute $\mathbf{u}_i$ via $\mathbf{u}_i = (1/\sigma_i)\, A\mathbf{v}_i$ (Fast-Track S3).

⚠ Trap 5: Computing the Larger Gram Matrix by Habit

The Mistake: Blindly computing A^TA for a 2×3 matrix, ending up with a 3×3 characteristic polynomial.

Rule: Always compute the smaller matrix (Fast-Track S1). For 2×3, compute AA^T (2×2 quadratic equation) to obtain non-zero singular values instantly.